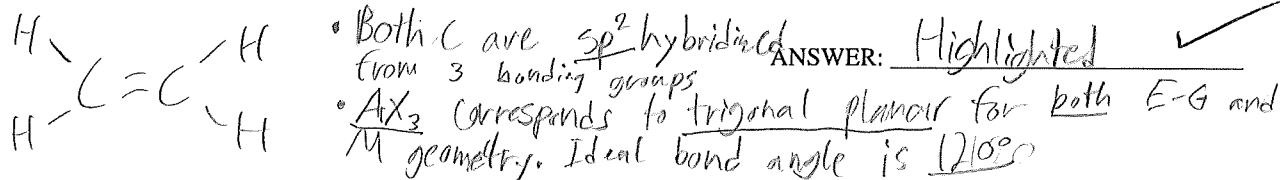


0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7	7.5	8	8.5	9	9.5	10	10.5	11	11.5	12	12.5	13	13.5	14	14.5	15
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Name: _____
These questions assess your understanding of the material from Chapters 7, 8 of OpenStax Chemistry. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

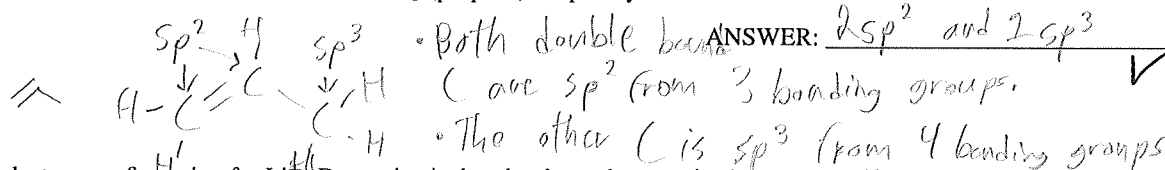
1. For the central atom in C_2H_4 , determine its hybridization, electron-group geometry, molecular geometry, and the ideal bond angle(s). Justify your answer.



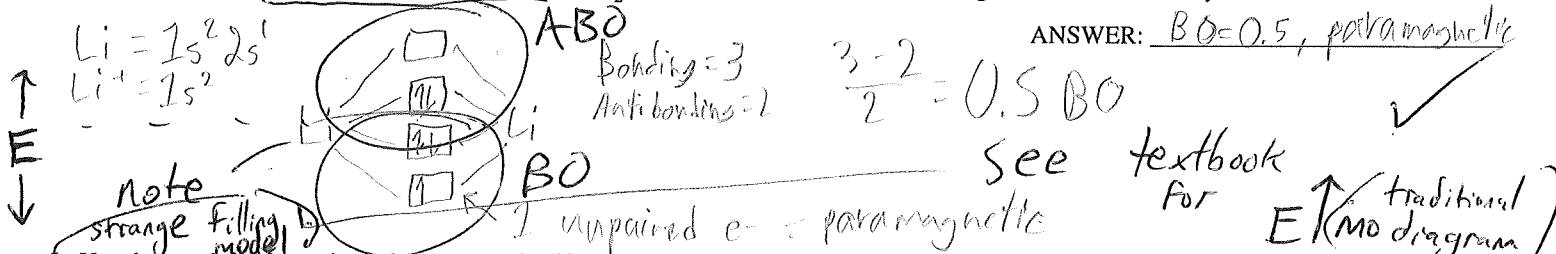
2. Which atom has the higher first ionization energy: magnesium (Mg) or sodium (Na)? You must briefly explain why. Credit is awarded only for explaining the cause of the trend based on atomic structure, not for merely stating the trend.

Both atoms experience shielding from core electrons and readily donate valence electrons. However, Mg's additional protons in the nucleus mean that the valence electrons are bound slightly tighter.
 ANSWER: Mg ✓

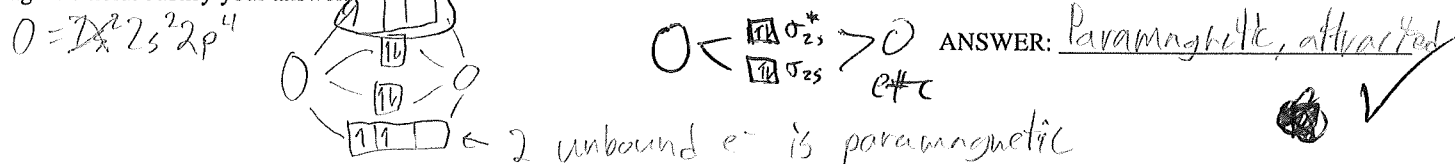
3. Identify the hybridization of each carbon atom in $CH_2=CH-CH_3$ (propene). Explain your answer.



4. Write the complete MO electron configuration for Li_2 . Determine its bond order and magnetic character. Justify each answer.



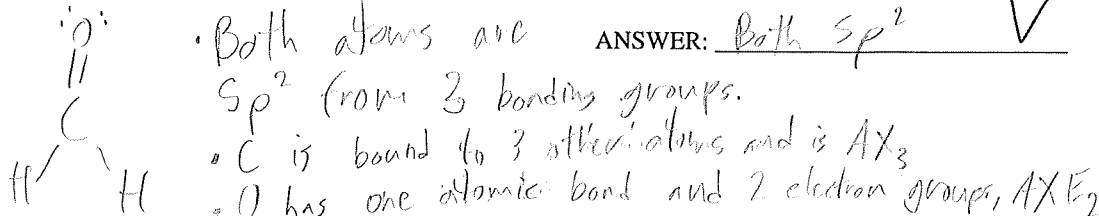
5. Use the number of unpaired electrons to classify O_2 as paramagnetic or diamagnetic, and state whether it is attracted to or repelled by a magnetic field. Justify your answer.



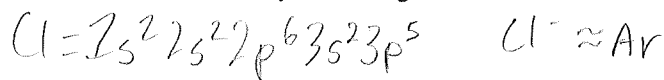
6. On the potential-energy curve for O_2 , what happens to the potential energy when the two atoms are very far apart? Why does this happen? Justify your answer.

When too far apart, O cannot effectively bond with the other O, and is not in its lowest potential energy state.
 ANSWER: Elevates (as they get farther apart) (but plateaus) ✓

7. Determine the hybridization of each atom (besides hydrogen) in CH_2O . Briefly justify each assignment using the local bonding environment of that atom.



8. Write the full electron configuration of a neutral chlorine atom. Then, using the octet rule, predict the charge of the monatomic ion that chlorine forms and identify the noble gas with which that ion is isoelectronic. Explain your reasoning.



ANSWER: -1, Ar ✓

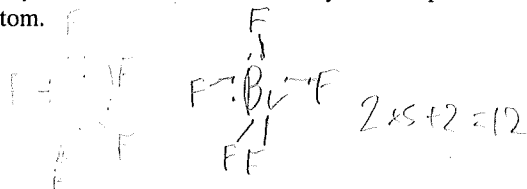
Cl forms the -1 charge, as gaining one would fill its octet and resemble Ar.

9. An atom forms both an sp hybrid orbital and an sp^3 hybrid orbital. Without calculating any numerical energies, determine which hybrid orbital has the higher expectation value of energy and justify your answer.

ANSWER: sp^3 ✓

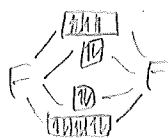
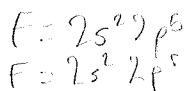
- s orbitals are lower energy than p.
- An sp orbital is half s, with some s character
- sp^3 orbitals are only a quarter s, with mostly p character
- More s character lowers energy, so sp is lower than sp^3

10. Does the central atom in BrF_5 obey the octet rule? If not, classify the exception as electron-deficient or expanded octet, and state the number of electrons around the central atom.



ANSWER: Expanded, 12 ✓

11. Use molecular-orbital theory to determine the bond order and whether F_2 is paramagnetic or diamagnetic. Justify your answer.



$BO = \frac{10 - 8}{2} = 1$

2 unpaired e⁻ = paramagnetic

ANSWER: $BO = 1$, paramagnetic

1/2

12. Determine the electron-group arrangement, molecular shape, and ideal bond angle(s) for methane.



• $C = AX_4$

• No lone pairs means that EG and M geometries are the same.

• $AX_4 = \text{tetrahedral} = 109.5^\circ$

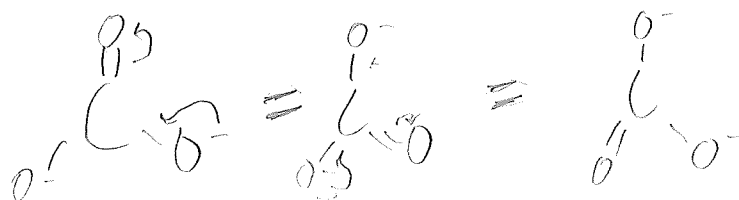
ANSWER: Tetrahedral ✓

13. Draw the Lewis electron-dot symbol for an atom of Ge.



ANSWER: Below ✓

14. Two students draw different valid Lewis structures for CO_3^{2-} . One assigns the central C as sp^2 ; the other claims the assignment depends on which Lewis structure is used. Which student is correct? Justify your answer. Justify your answer completely.



ANSWER: C is sp^2 ✓

good

C will have 3 bonding groups regardless of resonance form, and remains sp^2 . +6.5

15. Two molecules differ only in the molecular orbital occupied by their highest-energy electron. One molecule has a stronger bond than the other. Which molecule has greater electron density between the bonded nuclei in that occupied molecular orbital? Justify your answer.

ANSWER: The one with the stronger bond.

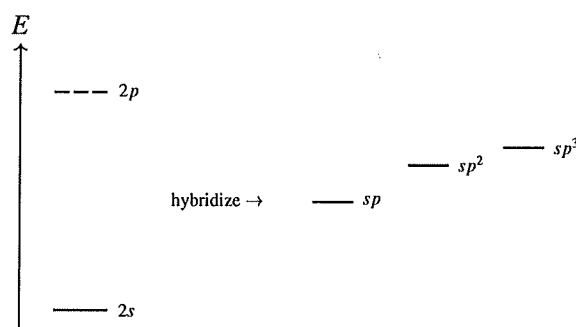
$sp^2 > sp^3$
 $sp > sp^2 \& sp^3$



↓
 • Stronger bond
 • Greater e^- density between orbitals

+

- sp^2 ; trigonal planar electron-group geometry; trigonal planar molecular geometry; $\approx 120^\circ$ $\therefore$ the Lewis structure contains 3 electron groups around the central atom. 3 electron groups require a trigonal planar arrangement, so the central atom uses sp^2 hybrid orbitals to orient those electron groups for minimum repulsion. The arrangement of bonding pairs and lone pairs within that electron-group geometry produces the observed molecular geometry and corresponding bond angle(s).
- Mg $\therefore$ effective nuclear charge increases across a period, so valence electrons are held more strongly
- sp^2 for C1 and C2 (alkene carbons); sp^3 for C3 (methyl carbon). C1 and C2 each have 3 electron groups (part of C=C), so sp^2 . C3 has 4 single bonds, so sp^3 $\therefore$ electron groups determine hybridization.
- MO configuration: $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^1$
Bond order = $\frac{3-2}{2} = 0.5$ $\therefore$ bond order is determined by the balance between electrons in bonding (3) and antibonding (2) molecular orbitals. Bonding orbitals stabilize the molecule, whereas antibonding orbitals destabilize it, so a larger excess of bonding electrons produces a stronger bond.
 Li_2^+ is paramagnetic $\therefore$ its magnetic character is determined by whether any molecular orbitals contain unpaired electrons. One or more unpaired electrons produces paramagnetism, whereas complete electron pairing produces diamagnetism.
- O_2 is paramagnetic and weakly attracted $\therefore$ its 2 unpaired electrons give a net magnetic moment that aligns with the field.
- Approximately zero $\therefore$ at very large separation the atoms are essentially noninteracting, and this state is defined as the zero-potential-energy reference.
- C: sp^2 — justified by 2 C–H bonds and 1 C=O bond
O: sp^2 — justified by 1 C=O bond and 2 lone pairs
- $1s^2 2s^2 2p^6 3s^2 3p^5$; Cl forms Cl^- and is isoelectronic with Ar $\therefore$ given the neutral Cl atom's electron configuration, it gains 1 electron to match the noble-gas electron configuration of Ar.
- sp^3 $\therefore$ it contains the greater fraction of p character. Because the $2p$ orbitals are higher in energy than the $2s$ orbital, increasing the contribution of $2p$ orbitals raises the weighted-average expectation value, whereas increasing s character lowers it.



- Exception: expanded octet $\therefore$ the central Br atom holds 12 electrons (more than 8; Br lies in Period 3 or beyond)
- MO config: $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \sigma_{2p}^2 \pi_{2p}^2 \pi_{2p}^{*2} \pi_{2p}^{*2}$
Bond order = $\frac{8-6}{2} = 1$
 F_2 is diamagnetic. The bond order is 1 because there are 8 bonding and 6 antibonding electrons. It is diamagnetic because all electrons are paired.
- e^- -group arrangement: Tetrahedral
Shape: Tetrahedral
Ideal bond angle(s): 109.5°
- $\cdot\ddot{Ge}\cdot$
-
- The student assigning sp^2 is correct. Every valid Lewis structure for this species has the same number of electron domains around the central atom (3), so the σ framework and electron-domain geometry are unchanged. Because hybridization follows from that geometry rather than the placement of π bonds, the assignment is identical for every valid Lewis structure.
- The molecule with the stronger bond has greater electron density between the bonded nuclei in that occupied molecular orbital $\therefore$ greater electron density between the nuclei produces stronger electrostatic attraction that holds the nuclei together more strongly.